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The Evolution of Statistical Induction Heads: In-Context Learning Markov Chains

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Abstract

Large language models have the ability to generate text that mimics patterns in their inputs. We introduce a simple Markov Chain sequence modeling task in order to study how this in-context learning (ICL) capability emerges. In our setting, each example is sampled from a Markov chain drawn from a prior distribution over Markov chains. Transformers trained on this task form *statistical induction heads* which compute accurate next-token probabilities given the bigram statistics of the context. During the course of training, models pass through multiple phases: after an initial stage in which predictions are uniform, they learn to sub-optimally predict using in-context single-token statistics (unigrams); then, there is a rapid phase transition to the correct in-context bigram solution. We conduct an empirical and theoretical investigation of this multi-phase process, showing how successful learning results from the interaction between the transformer's layers, and uncovering evidence that the presence of the simpler unigram solution may delay formation of the final bigram solution. We examine how learning is affected by varying the prior distribution over Markov chains, and consider the generalization of our in-context learning of Markov chains (ICL-MC) task to n-grams for $n > 2$.

1 Introduction

Large language models (LLMs) exhibit a remarkable ability to perform *in-context learning* (ICL): learning from patterns in their input context (Brown et al., 2020; Dong et al., 2022). The ability of LLMs to adaptively learn from context is profoundly useful, yet the underlying mechanisms of this emergent capability are not fully understood.

In an effort to better understand ICL, some recent works propose to study ICL in controlled synthetic settings—in particular, training transformers on mathematically defined tasks which require learning from the input context. For example, a recent line of works studies the ability of transformers to perform ICL of standard supervised learning problems such as linear regression (Garg et al., 2022; Akyürek et al., 2022; Li et al., 2023; Wu et al., 2023). Studying these well-understood synthetic learning tasks enables fine-grained control over the data distribution, allows for comparisons with established supervised learning algorithms, and facilitates the examination of the in-context "algorithm" implemented by the network. That said, these supervised settings are reflective specifically of *few-shot learning*, which is only a special case of the more general phenomenon of networks incorporating patterns from their context into their predictions. A few recent works (Bietti et al., 2023; Xie et al., 2022) go beyond the case of cleanly separated in-context inputs and outputs, studying in-context learning on distributions based on discrete stochastic processes.

*Work done while visiting Harvard University.

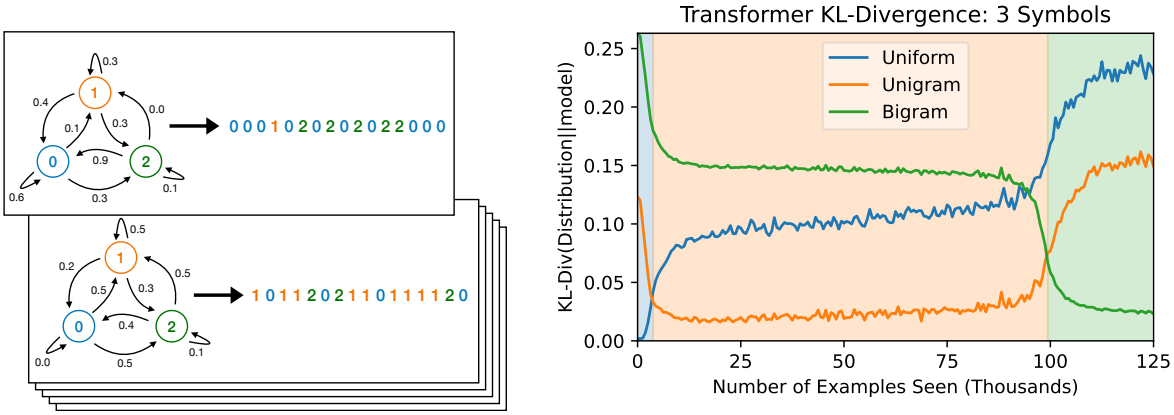

Figure 1: (**left**) We train small transformers to perform in-context learning of Markov chains (ICL-MC)—next-token prediction on the outputs of Markov chains. Each training sequence is generated by sampling a transition matrix from a prior distribution, and then sampling a sequence from this Markov chain. (**right**) Distance of a transformer's output distribution to several well-defined strategies over the course of training on our in-context Markov chain task. The model passes through three stages: (1) predicting a uniform distribution, (2) predicting based on in-context unigram statistics, (3) predicting based on in-context bigram statistics. Shading is based on the minimum of the curves.

The goal of this work is to propose and analyze a simple synthetic setting for studying ICL. To achieve this, we consider n-gram models (Brown et al., 1992; Shannon, 1948; Chomsky, 1956), one of the simplest and oldest methods for language modeling. An n-gram language model predicts the probability of a token based on the preceding $n-1$ tokens, using fixed-size chunks (n-grams) of text data to capture linguistic patterns. Our work studies ICL of n-gram models, where the network needs to compute the conditional probability of the next token based on the statistics of the tokens observed in the input context, rather than on the statistics of the entire training data. We mainly focus on the simple case of $n = 2$; i.e., bigram models, which can be represented as Markov chains. We therefore consider ICL of Markov chains (ICL-MC): we train a transformer on sequences of tokens, where each sequence is produced by a different Markov chain, generated using a different transition matrix (see Figure 1 (left)).

By studying ICL-MC, we are able to replicate and study multiple phenomena that have been observed in ICL for LLMs, and identify new ones. We demonstrate our findings using a combination of empirical observations on transformers trained from scratch on ICL-MC and theoretical analysis of a simplified linear transformer. Our key findings are summarized below:

- **Transformers learn statistical induction heads to optimally solve ICL-MC**. Prior work studying ICL in transformers revealed the formation of *induction heads* (Elhage et al., 2021), a circuit that looks for recent occurrence(s) of the current token, and boosts the probabilities of tokens which followed in the input context. We show that in order to solve ICL-MC, transformers learn *statistical* induction heads that are able to compute the correct *conditional (posterior) probability* of the next token given all previous occurrences of the prior token (see the attention patterns in Figure 2). We show that these statistical induction heads lead to the transformers achieving performance approaching that of the Bayes-optimal predictor.

- **Transformers learn predictors of increasing complexity and undergo a phase transition when increasing complexity.** We observe that transformers display *phase transitions* when learning Markov chains—learning appears to be separated into phases, with fast drops in loss between the phases. We are able to show that different phases correspond to learning models of increased complexity—unigrams, then bigrams (see Figure 1)—and characterize the transition between the phases. We also consider the n-gram generalization of our setting where the next token is generated based on the previous $n-1$ tokens.

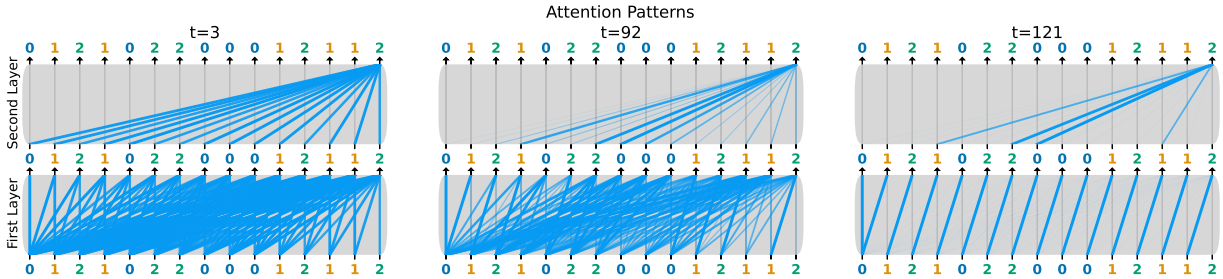

Figure 2: Attention for a fixed input at various time steps in training. These diagrams show where the attention heads are attending to at each layer. In the second layer, only the last token attention is shown. Tokens on top attend to tokens below them. Attention starts off uniform, but by the end of training, the layers are clearly acting the same as the induction head construction. Specifically, in the first layer each token is attending to the previous token. In the second layer, the current token, a 2, is attending to tokens that followed 2s, allowing bigram statistics to be calculated. Figure 10 shows the full attention matrices as heatmaps.

- **Simplicity bias may slow down learning.** We provide evidence that the model's inherent bias towards simpler solutions (in particular, in-context unigrams) causes learning of the optimal solution to be delayed. Changing the distribution of the in-context examples to remove the usefulness of in-context unigrams leads to faster convergence, even when evaluated on the original distribution.
- **Alignment of layers is crucial.** We show that the transition from a phase of learning the simple-but-inadequate solution to the complex-and-correct solution happens due to an alignment between the layers of the model: the learning signal for the first layer is tied to the extent to which the second layer approaches its correct weights.
- **Alternating patterns in positional embeddings.** When we train transformers with relative position embeddings, the theoretical optimization analysis of our simplified model indicates that along the way to the correct solution, the first layer develops a bias toward looking back an *odd* number of tokens, even though only looking back by 1 is clearly useful. We empirically observe this curious phenomenon in real transformers as well.

1.1 Related Work

In-context Learning. Recently, many works have focused on understanding how ICL emerges in language models. In (Chan et al., 2022), the authors discuss how properties of the data distribution promote ICL, with a focus on empirical observations. Xie et al. (2022) studies a data model similar to ours, demonstrating that language models trained on Hidden Markov Models (HMMs) can learn in-context HMMs not found in the training data. Abernethy et al. (2023) study the ability of transformers to segment the context into pairs of examples and labels and provide learning guarantees when the labeling is of the form of a sparse function. The work of Bietti et al. (2023) studies the dynamics of training transformers on a task that is reminiscent of our Markov chain setting but has additional complexities. Instead of drawing a fresh Markov chain for each sequence, in their task all sequences are sampled from the same Markov chain; after certain 'trigger' tokens, the following 'output' token is chosen deterministically within a sequence. Thus, successful prediction requires incorporating both global bigram statistics and in-context deterministic bigram copying, unlike in our setting where the patterns computed by *statistical* induction heads are necessary and sufficient. As in our work, they identify multiple distinct stages of training and show how multiple top-down gradient steps lead to a solution.

Other works observe that ICL is possible due to the ability of transformers to implement gradient descent as a "meta learning" algorithm, and show some evidence that this indeed corresponds to how transformers